

NIMBYism and the Housing Supply*

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Abstract

We develop a political economy model of the housing market in which the supply of new homes is shaped by a democratic vote among households. Voter support for additional housing is determined by the effect this vote has on households' expected discounted utility via changes in house prices. Households that own housing, and thus will often be made poorer by additional supply, generally oppose new housing – the “Not In My Backyard” or NIMBY effect. Households that wish to buy a house and would benefit from cheaper housing support more supply. We find that support for additional supply is highest among households that are younger, low-income or are currently renting. We then simulate the impact of a temporary baby boom in the population. We find that those born during the boom have higher rates of homeownership and higher net worth due to their disproportionate voting power. However, this comes at the expense of higher house prices, lower homeownership rates and less wealth for the subsequent generations. Calibrating the model to US demographic data, we find this political economy mechanism was responsible for a 15 percent increase in house prices since 1950. Using demographic projections we forecast that due to population aging house prices will continue to rise into the 21st century.

JEL Classification: E24, J10, R31.

Keywords: Housing supply, inequality, political economy

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1 Introduction

The “Not In My Backyard” (NIMBY) effect describes the phenomenon of local residents opposing new housing developments in their neighborhood. As the regulation of housing supply is often decentralized to local governments, residents can exert significant political power over any new developments within their area. The aggregate effect of NIMBYism can thus negatively impact the housing supply within an economy. In the US, for example, land is divided into “zones” which determine the size and use of new developments.¹ Only detached, single-family homes are allowed to be built in “R1” residential zones, even if high demand for multi-home housing (such as apartments) exists. Those who own homes within R1 residential areas have strong incentives to maintain the status quo at the expense of non-homeowners, who would benefit from multi-home housing in order to get on the housing ladder (Manville et al. 2020).

We develop a political economy model of the housing market in order to estimate the impact of NIMBYism and show how changing demographic trends can explain the recent rise in the house prices. Specifically, we allow housing supply in a standard life cycle model to be determined by a democratic process: households vote according to their personal preferences and the final outcome is determined by the median voter. We then use this model to show that an aging society exacerbates the effects of NIMBYism, and has led to a 15 percent increase in the house prices since 1950. Using demographic projections we forecast population aging will continue to increase house prices in the 21st century. Finally we examine the dynamics of a temporary baby boom and analyze how it affects the level and distribution of wealth in the economy. We find that the generation born during the baby boom will collectively use their increased voting power to expand their ownership of housing and increase their total net worth. However this comes at the cost of decreased home-ownership rates and networth for subsequent generations.

Our theoretical model follows the standard life cycle housing model as described by Iacoviello (2005). Over the life-cycle, an individual household typically rents housing until they have saved enough to meet their borrowing constraint and be able to put a deposit down on a house. Households also participate in the housing market by either upsizing (as their wealth grows or positive income shocks occur) or selling (if they receive a negative income shock or die). Instead of fixing the housing supply (as in Iacoviello 2005) or having it produced inelastically by a construction sector (as in Kaplan et al. 2020), we allow the supply of housing to be determined by a democratic vote of local households.

Household preferences over the supply of housing are formed according to the impact that a small change in price will have on their expected discounted utility. Households who gain from marginally higher prices will vote against new developments, whereas those who marginally lose out from higher house prices will vote in favor of new developments. NIMBYist views are thus

¹Zoning laws were originally developed in the 1900s in order to improve “light and air” in residential areas - see Fischer (2002) for a historical overview. Land use regulation has also been used by white residents within the US to maintain and increase racial segregation (Sahn 2021, Trounstein 2020).

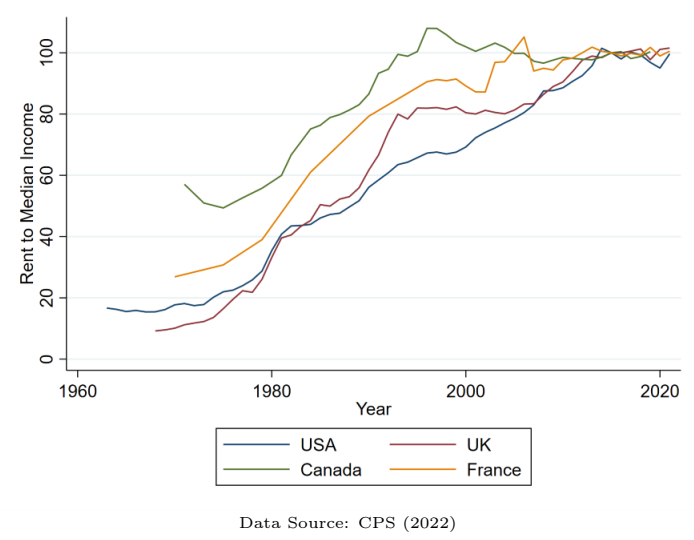
largely determined by households' ownership of housing and their expected probability of being a net-buyer in future periods. NIMBYist households are disproportionately older, wealthier, have higher incomes and are more likely to be homeowners, which is in line with the empirical literature surrounding NIMBYism (see Whittlemore & BenDor (2018), Scally and Tighe (2015), and Hankinson (2018) for a range of surveys on the subject).

Our model primarily focuses on the financial incentives of NIMBYism, as households' own homes are the primary asset for most households.² However there are several non-financial motivations that may lead a household to oppose new developments - ranging from infrastructure capacity concerns to simply disliking the proposed style of architecture or neighborhood characteristics (Ahlfeldt 2011, Bosetti and Sims 2016). We allow for these non-financial preferences to affect households' votes but assume these preferences are stable over the life cycle. The financial incentive for NIMBYism, however, does vary over the life cycle and is analogous to an insider-outsider problem: outsiders (renters) would prefer more houses to be built until they become an insider (homeowner), and then seek to restrict supply.

As homeownership rates and thus support for new developments changes over the life cycle, the demographic make-up of an economy will play an important role in the aggregate effect of NIMBYism. During the mid 1940s, a baby boom occurred in a number of OECD countries including the US, UK, New Zealand, Australia, France and Canada (Greenwood et al. 2005). This baby boom lasted until the subsequent baby bust of the mid 1960s when birth rates returned to trend. The effect of the spike in the birth rate led to a generation known as the Baby Boomers which has taken up a significant share of the population over time. As those who are close in age will have similar preferences over housing on average (as a result of being at a similar stage of the life-cycle), those born during the baby boom will have outsized political power and thus have a greater influence over the aggregate housing stock. This is perhaps one reason why rents are increasing relative to median income over time as shown in figure 1.

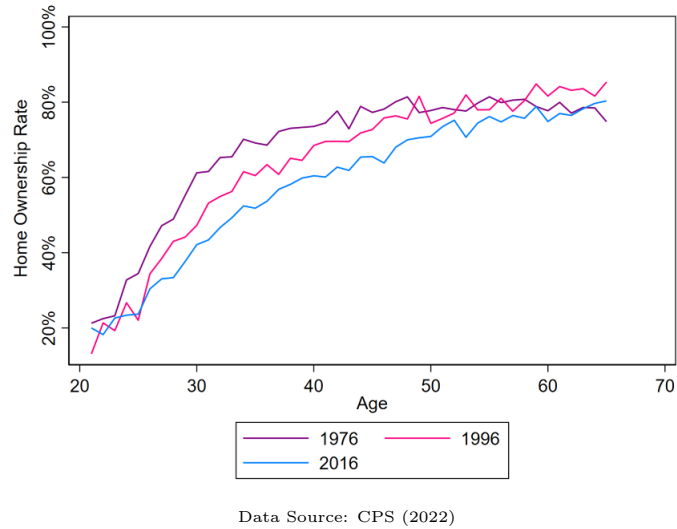
²Excluding the top 1% of the wealth distribution, home equity makes up 28.9% of total US household wealth and is comparable only to retirement accounts 32.8%. The next largest asset, stocks and mutual funds, makes up only 10.2% of total household wealth (see Eggleston et al. 2020).

Figure 1: Rent to median income ratio over time



It is also why the US homeownership rate has been falling for younger people over time as shown in figure 2.

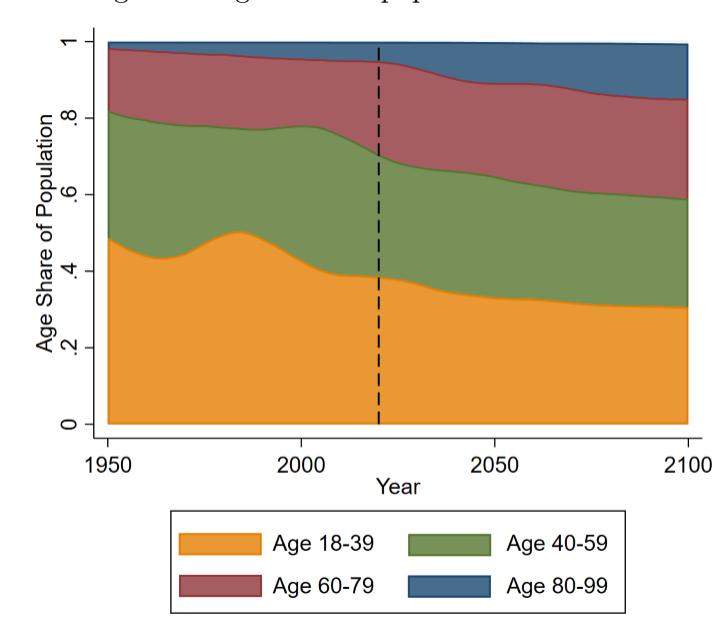
Figure 2: Home ownership rate for different age cohorts



In this paper, we estimate how the changing demographic shares across aging economies have changed the political-economic equilibrium in the housing market. As economies age, the balance of political power shifts in favor of NIMBYist households resulting in lower housing supply and higher house prices. We find that house prices increased by around 15 per cent due to these political economic effects since the 1950s. Using demographic projections, as shown in figure 3 we find that the aging economy is expected to keep the house price at or above it's current level untill 2100.

Finally, we use our political economic model of the housing market to examine the distributional consequences of recent demographic shocks. We find that a temporary boom in the birthrate will initially decrease inter and intra-generational wealth inequality as the boom generation initially supports lower house prices when they are disproportionately young renters. As the cohort ages, this effect reverses and this cohort greater political pressure against the supply of new housing. This decrease in the support for housing supply, and thus higher house price, leads to a permanent scarring effect on future generations who are unable to achieve home ownership at the same rate as previous generations, even after the initial baby boom generation has completely died off.

Figure 3: Age share of population over time



Data Source: UN (2019). Dashed line indicates population forecasts for year 2020 onward assuming mid level immigration

2 Related Literature

This paper is related to the following strands of the literature. Firstly, there is growing literature surrounding the impact of regulation on the housing supply. The concept of NIMBYism is often attributed to Frieden (1979), who noted a rise in environmental regulations to impede new housing developments after the 1970s. Gyourko et al. (2013) support this view and argue that before the 1970s, housing construction tended to follow areas of high price growth. However, establishing a direct link between regulation and the housing supply is difficult due to the idiosyncratic forms that regulations can take in different localities (Gyourko and Malloy 2015). To get round this, Glaeser et al. (2006) offer indirect evidence of excess regulation by noting the existence of a relatively large wedge between prices and marginal constructions costs. The

difficulty in providing direct evidence stems from the endogeneity between land use regulation and the housing market. Recent papers have used boundary discontinuity design to address this issue (Kahn et al. 2010, Turner et al. 2014, Kulka, 2019). For example, Anagol et al. (2021) exploit a zoning reform in Sao Paulo and show that housing developers apply for more permits in blocks with higher maximum densities. In their equilibrium model of housing supply this leads to a 2.2% increase in the total housing stock and 0.5% average price reduction. Similarly, Song (2021) constructs a nationwide data set of minimum lot area estimates and finds that a doubling the US minimum lot area size would increase house prices by 14% and rents by 6%.

This paper is also related to an older strand of literature on political redistribution such as: Alesina and Rodrik (1994), Persson and Tabellini (1994), Krussel and Rios-Rull (1999), Acemoglu and Robinson (2001). The main idea of these papers is that when income and resources are distributed heterogeneously, the majority may vote for redistribute policies. These papers rely on a voting mechanism, the median-voter theorem, to determine what policy will be pursued. The theorem shows that when voting takes place over a single issue, and preferences are singled peaked, then the policy chosen under a democracy will be the one chosen by the median-voter.

Housing politics is a good candidate for the median-voter theorem as voting take place at local level and usually on specific propositions related to relaxing zoning regulations. The idea of homeowners as specific voting class in local government decisions as a result of financial interests is related to the “homevoter hypothesis” (Fischel 1999, 2001). Einstein et al. (2019) examine planning meetings in Massachusetts and find that homeowners and older residents are overly representative in planning meetings. This is in line with voter turnout in local elections which in general tend to be from these two groups (Kogan et al. 2018, Oliver and Ha 2007). Similarly, Jiang (2018) finds that voter turnout increase for mayoral elections, relative to presidential elections, when home-ownership rates increase. In addition, Cheung and Meltzer (2013) find that the prevalence of Homeowners Associations is positively associated with an increase in land use regulations. Similarly, Hilber and Robert-Nicoud (2013) find a positive correlation between the degree of physical development in an area and land use regulation. Both Ahlfeldt and Maennig (2015) as well as Dehring and Ward (2008) use public referendums to directly test the homevoter hypothesis. Ahlfeldt and Maennig (2015) use a referendum on the expansion of an airport in Berlin as a natural experiment and find greater support among homeowners as opposed to leaseholders. The support for the proposal comes from two effects: a consumption effects (though use of the amenity) as well as a wealth effect (via house price increases). Dehring and Ward (2008) use pre-vote changes in residential property values to test the probability of support for the new stadium for the Dallas Cowboys and find between a 0.9% and 1.2% increase in support for every \$1000 dollar increase in house prices.

Numerous surveys of voters have attempted to describe the demographics of voters who oppose new housing supply. Whittlemore and BenDor (2019) find that older, less educated were more likely to oppose new developments, though they found no significant relationship between

home ownership and NIMBYist views. By contrast Hankinson (2018) and Scally and Tighe (2015) found that home owners were significantly more likely to oppose new housing supply - particularly in their locality.

Finally, this paper is related to the literature surrounding life-cycle models with housing markets. Generally, life-cycle models assume exogenous house prices due to computational complexity (Fernandez-Villaverde and Krueger (2011), Iacoviello and Pavan (2013), Yang (2009)). There are, however, relatively few papers where aggregate shocks affect house prices (Justiniano et al. 2015). Our paper is most closely related to Ortalo-Magné and Prat (2014), who use a median-voter theorem in an overlapping-generations economy to show that increasing homeownership rates decrease housing affordability. Similarly, Mankiw and Weil (1989) predict the growth rate of housing demand via changes in the age distribution alone and more recently Mumtaz and Šustek (2023) find similar results except housing demand peaks in the 2000s.

3 The Model

In this section we outline a life-cycle model with housing and debt choices. The model features heterogeneous, life-cycle households that optimize their consumption of non-durable consumption, housing and saving in liquid assets. A number of factors influence households preferences on house prices and thus their political support for additional housing supply. These include household income and wealth as well as portfolio allocation and age over the life cycle. We use the model to study the housing market response to an exogenous change in demographics. Following the shock, the behavior of owner-occupiers and renters determines equilibrium house prices.

A consumer lives for T periods and in each period for, $t \leq T$ maximizes lifetime expected utility which consists of consumption c_t and housing services, h_t . Housing services depends on whether they own or rent the property. Individuals also have access to a liquid asset that pays an interest r_{t+1} . The household's lifetime utility function is given by

$$U = \sum_{j=0}^J \beta^{j-1} u(c_j, h_j) + \beta^J v(w_{J+1}) \quad (1)$$

where households optimize over consumption, c_t , liquid assets, a_t , and housing, h_t at each age j . The flow utility is increasing and strictly concave in c and h . Households receive utility from wealth they bequest w_{J+1} . We assume the functional form

$$u(c_t, h_t) = \frac{(c_t^{1-\zeta} h_t^\zeta)^{1-\gamma}}{1-\gamma} \quad (2)$$

where the consumption of housing stock consists of either the amount of housing owned or rented

$$h_t = \begin{cases} \theta^{rent} h_t^{rent} & \text{if } h_t^{own} = 0 \\ h_t^{own} & \text{if } h_t^{own} > 0. \end{cases} \quad (3)$$

The curvature parameter γ determines households' risk aversion and inter-temporal elasticity of substitution and ζ denotes housing's share of total consumption. Finally there is a warm glow bequest function when households die that provides utility

$$v(w) = \theta \frac{(w + \bar{w})^{1-\gamma}}{1-\gamma} \quad (4)$$

which generates a bequest function with decreasing marginal utility with respect to wealth governed by the parameters $\bar{w}$ and θ . Households supply their labor inelastically and are paid a wage $y_{t,j}$. The wage they receive varies according to their productivity which is driven by the exogenous process

$$\ln y_{t,j} = g_j + z_t \quad (5)$$

where z_t is a Markov process calibrated to yield realistic inequality (as per Kindermann and Krueger (2022)) and g_j is a deterministic age profile. The stochastic process comprises of an AR(1) process for lowest income states, with two additional top-income states which incorporate "superstar" incomes. Once workers reach retirement age J_{ret} income falls to a fixed level based on their wage upon retirement. The income process is normalized to have a mean of 1, such that the price of housing p_t^h reflects the house price to income ratio.

Households may save or borrow liquid assets subject to a borrowing constraint where each household can borrow against a proportion, m , of the total value of the stock of housing they own

$$a_t \geq -m h_t^{own} p_t^h. \quad (6)$$

Households may buy or sell housing at price, p_t^h , subject to a minimum block size $\bar{h}$. All property sales, including those that are sold as part of a bequest are subject to a transaction cost F_s proportional to the total value of property sold. If they do not own any housing they must rent it from real estate firms at price p_t^{rent} . The household's utility function is thus maximized subject to the budget constraint

$$a_{t+1} = (1 + r_t^a) a_t + y_t - c_t - p_t^{rent} h_t^{rent} + I^{sell} (1 - F_s) p_t^h \Delta h_t^{own} \quad (7)$$

where I^{sell} is an indicator variable for households that choose to sell part or all of their current stock of housing.

3.1 Household's Problem

Each household's state vector is $\{a_t, h_{t-1}, z_t\}$, where a_t is the stock of liquid assets, h_{t-1} is the amount of housing they started the period with and z_t is the persistent component of their income. The value function for the household is given by the maximum of the three possible housing tenure options: rent, adjust their stock of housing or maintain their current level of housing. The value function for renters is

$$\begin{aligned}
 V_j^{rent}(c_t, h_t^{own}, z_t) &= \max_{c_t, a_t, h_t^{rent}} u(c_t, h_t^{rent}) + \beta EV_{j+1}(c_{t+1}, h_{t+1}^{own}, z_{t+1}) \\
 &\quad s.t \\
 a_{t+1} &= (1 + r_t^a) a_t + y_t - c_t - p_t^{rent} h_t^{rent} + (1 - F_s) p_t^h h_t \\
 a_{t+1} &\geq 0 \\
 h_{t+1} &= 0
 \end{aligned} \tag{8}$$

The value function for households who choose not to adjust is

$$\begin{aligned}
 V_j^{non-adjusters}(a_t, h_t^{own}, z_t) &= \max_{c_t, a_t} u(c_t, h_t) + \beta EV_{j+1}(a_{t+1}, h_{t+1}^{own}, z_{t+1}) \\
 &\quad s.t \\
 a_{t+1} &= (1 + r_t^a) a_t + y_t - c_t \\
 a_{t+1} &\geq -mp_t h_t^{own} \\
 h_t^{own} &= h_{t+1}^{own}
 \end{aligned} \tag{9}$$

The value function for households who choose to buy or sell housing is

$$\begin{aligned}
 V_j^{adjust}(a_t, h_t^{own}, z_t) &= \max_{c_t, a_t, h_t^r} u(c_t, h_t^{own}) + \beta EV_{j+1}(a_{t+1}, h_{t+1}^{own}, z_{t+1}) \\
 &\quad s.t \\
 a_{t+1} &= (1 + r_t^a) a_t + y_t - c_t + I^{sell} (1 - F_s) p_t^h \Delta h_t^{own} - I^{buy} (1 - F_b) p_t^h \Delta h_t^{own} \\
 a_t &\geq -mp_t h_t^{own} \\
 h_{t+1}^{own} &= h_t^{own} + \Delta h_t^{own}
 \end{aligned} \tag{10}$$

The final value function is the maximum of each of the three choices over housing tenure

$$V_j(a_t, h_t^{own}, z_t) = \max \left\{ V_j^{rent}, V_j^{adjusters}, V_j^{non-adjusters} \right\}. \tag{11}$$

3.2 Real Estate Firms

Real estate firms borrow from households to fund the purchase of housing, which is rented out to households that do not own their own home. Real estate firms operate in a perfectly competitive market, and thus the rental price is pinned down by the cost of borrowing, a fixed cost of providing rental services, the price of purchasing house stock and any expected capital appreciation driven by the drift in the price of housing. We assume that real estate firms are able to borrow against the full value of their housing stock. The zero profit condition pins down the price of rental stock as the function of the cost of housing, capital gains from housing and the interest rate

$$p_t^{rent} = \phi^{rent} + (r_t) p_t^a - \Delta p_t^a, \quad (12)$$

where ϕ^{rent} is the fixed cost of providing rental services.

3.3 Housing Supply

The aggregate economy consists of unit measure of neighborhoods across which all types of households are equally distributed. The neighborhood's housing supply in period $t + 1$ is determined in each neighborhood by a simple majoritarian voting mechanism in which households vote over the amount of permitted housing in an area. Only households who currently reside in that neighborhood can vote to determine the next periods level of new housing. An increase in the housing supply will reduce the price of housing in the locality and households can vote assuming, correctly, that an increase in housing supply will lead to a small persistent decrease in the cost of housing. An increase in the housing supply will also affect the price of rental properties by lowering the input price for the real estate firms. Each household in a neighborhood will vote to increase (or decrease) the availability of housing in their locality based on how a small, but permanent, change in house prices will effect their expected welfare. Specifically, households will vote to increase or decrease the supply of housing if a change in house prices increases their expected lifetime utility such that

$$\frac{\partial V_j(a_t, h_t^{own}, z_t)}{\partial p_t} + \sigma \geq 0 \quad (13)$$

where σ represents the impact of a non-financial preferences over additional housing supply. These non-financial preference over the optimal house price include both ideological preferences over the desired supply of housing as well as the value (or cost) of any change in amenity caused by additional supply of housing.³

Applying the median voter theorem, equilibrium in the housing market will occur when the median voter is indifferent to a change in the price of housing.⁴ A positive (negative) σ indicates

³We do not take a strong stance on the sign of these non-financial preferences, but given they include both potential costs of additional density (i.e. from congestion) and benefits (from agglomeration) it is plausible they could be positive or negative.

⁴While these preferences are in general well defined, it is important to note that preferences over house prices

an ideological preference for higher housing supply and lower house prices (lower housing supply and higher house prices).

The partial derivative of expected lifetime utility with respect to the price housing is determined by the households current portfolio of assets and expected life-time income. Figure 4 shows how different household characteristics correlate with the decision to vote for lower supply and thus higher house prices. Households who currently rent or expect to have a high likelihood of purchasing new housing in the near term will vote for housing supply to increase thus decreasing the price of housing and relaxing their budget constraint. These households tend to be younger, have low incomes and low wealth. Looking at the distribution of voters according to the expected probability that a given household will make a purchase of housing at some point in their life, we can see that households that do not have a high expected probability of purchasing housing over the rest of their life-time will generally prefer a higher price of housing in order to increase the size of their bequest and the resale value of their stock of housing. By contrast the coalition of support for lower house prices consists of two distinct groups of households: households that anticipate buying additional housing (particularly in the near term) and thus want to loosen their budget constraint. as well as households that are currently renters because an increase in the supply of housing also lowers the price of rental units.

The equilibrium price of housing will thus vary according to the relative size of these groups of households. An exogenous increase in young households, who disproportionately rent and thus prefer lower house prices, will lead to an increase in the supply of housing. The price of housing will thus fall until a sufficient number of renters become homeowners, and accordingly switch their political support towards higher house prices, thus making the median household indifferent to a change in house prices.

In order to abstract from strategic voting across dynasties, we assume different children and parents live in different neighborhoods.⁵ Consequently even if an individual may gain a larger inheritance if the price of their parents home increases, it will not effect their vote on the optimal level of housing supply in their local neighborhood.⁶

We assume that households are unable to forecast future changes in political support for additional housing supply and instead assume that the house prices follow a random walk. This ensures the dynamics of the model are solvable as it does not require the households to consider the possibility of strategic voting (i.e. voting to lower the house price today to ensure wider

are not monotonic. For example, a household with a moderate amount of housing might prefer a small increase in house prices as this will increase their expected bequest, but would be better off if house prices fall substantially as they would be able to consume more housing.

⁵According to Fu (2019) 61.5% adults live more than 10 miles away from their mother.

⁶This assumption also makes our model robust to other forms of altruistic giving. For example, in addition to receiving a warm glow from giving to their descendants at death, households may care about their descendant's welfare directly and thus may wish to sell housing in order to give their children funds to purchase housing earlier. As families live in different neighbourhoods, however, this would still not change their voting behaviour. This is because parents with housing would still wish to restrict housing supply in their neighbourhood in order to receive the highest return on their house so they can maximise their child's welfare.

support for higher house prices in future periods). We also assume that households are unable to form coalitions when voting, nor bargain political support across time periods and vote solely based on their individual interest at time t .

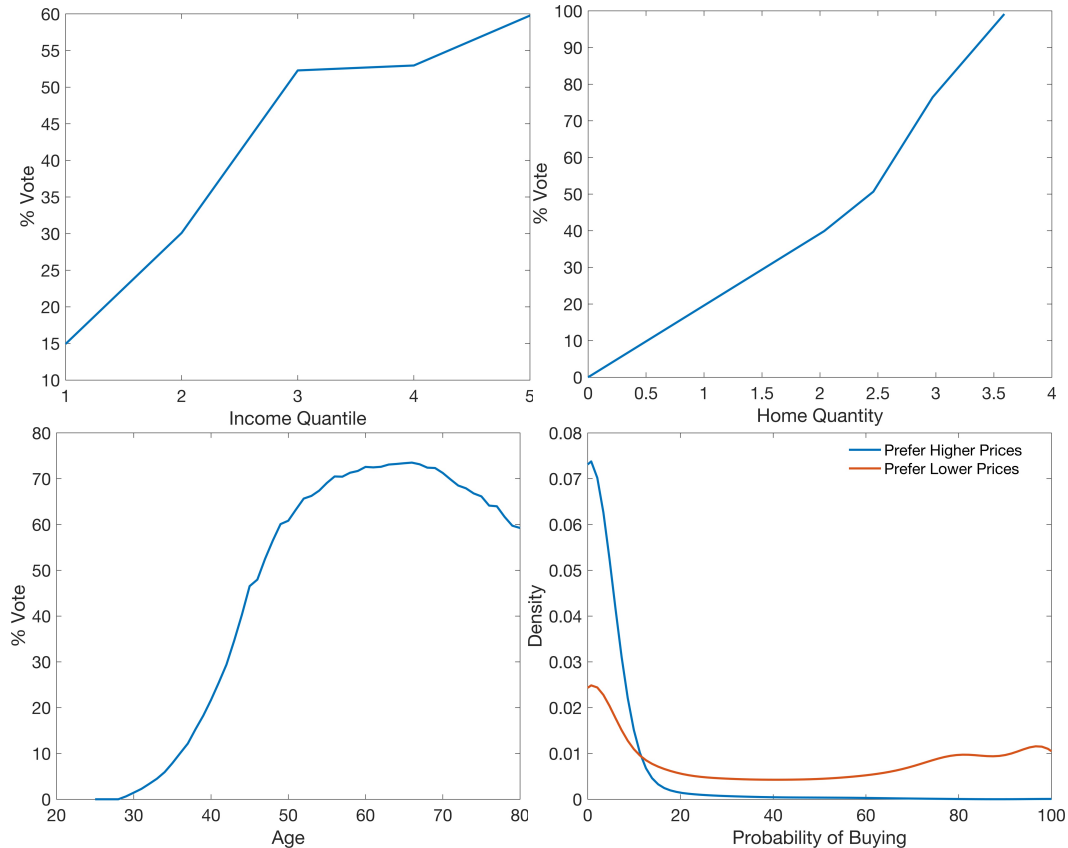


Figure 4: Household Support for Higher House Prices

4 Calibration

We calibrate the model to capture salient features of the US housing market in the early 2000s prior to the housing boom and bust. The model period is one year, households work for 41 periods (age 25 to 65) and die after 56 periods (age 80). The risk aversion parameter is set to 2, as is standard in the macroeconomics literature. The income process consists of the parameters for the deterministic age-profile, the persistent AR(1) component, and the transitory i.i.d. component of income. We estimate the parameters of the deterministic and stochastic 20 income processes using data from the Panel Study of Income Dynamics.

The retirement income replacement rate is set at 50% of the final period's non-transitory income, based on Díaz et al. (2008). The risk-free interest rate is set at 2%, while the mortgage

interest rate is 5%. The maximum loan-to-value (LTV) is set at 0.9 and 0.4 in line with Greenwald (2018).

Table 1: Parameter Values

Parameter	Value	Interpretation	Source
r^{spread}	3%	Spread between mortgage and saving rate	30 year average
J	56	Periods of life	
J_{ret}	41	Period of working life	
r^a	0.02	Risk free rate	30 year average
β	0.978	Discount rate	Internally Calibrated
γ	2	Risk aversion	Mean IES of 0.75
ζ	0.75	Non-durable share	PSID data
k_a	0.07	Cost of selling house	KMV (2017)
M	0.9	Maximum LTV	Guerrieri and Iacoviello (2017)
θ	10	Bequest scaling	Internally Calibrated
$\bar{w}$	150	Bequest parameter	Internally Calibrated
θ^{rent}	0.9	Rental discount	Mian et al (2017)

We set the house price, rental discount, discount rate, utility housing weight and real estate markup $\{p^h, \sigma^v, \bar{w}, \theta, \beta\}$ using simulated method moments to match the following moments.

Table 2: Calibration

	Model	Data
Share of owners with a mortgage	0.55	0.57
Share of renters	0.38	0.38
Home-ownership rate 25-35	0.38	0.40
Home-ownership rate 45-55	0.72	0.70
Home-ownership rate 65-75	0.71	0.75

The identity of the median voter will be defined by σ , the households ideological preference over the supply of housing, this allows households to have non-financial preferences over the supply of housing. We calibrate σ such that the housing market is in equilibrium in 2000. Our calibration requires a non-financial preference in favor of approving more homes as homeowners almost always make up a large majority of the voting population.

5 Quantitative Analysis

5.1 Steady State

First we analyze how demographic changes affect the housing market in the model's steady state. We calculate the political-economic equilibrium using demographic data from the US over the period 1950-2020. During this time there was substantial variation in the demographic structure - driven by the generation born in the post-war spike and how it reverberated over time - as shown in figure 3 at the beginning of the paper. For example, the mean age starts at 30 in 1950 and eventually falls as those born during the baby boom increases the share of young people in the labor and housing market. The mean age in the labor market troughs at 28 in 1970 as the baby boom subsides. The mean age then continues to steadily rise to 39 in 2020 as the those born during the baby boom ages.

To estimate the effect of the changing demographic composition for each year in this period, we solve for the equilibrium price which would result in the median voter being indifferent over an increase or decrease in housing supply. All other parameters are kept constant with the only change coming via variation in demographic structure.

Since homeownership rates increases with age, we find that when the median population age increases the aggregate homeownership rate rises. This increases political support for a reduction in the housing supply which increases house prices. House prices rise until there is enough political support for additional housing supply (via an increase in households who rent) such that the median voter is again indifferent to a small change in house prices.

Figure 5 shows the political-economic equilibrium house price over the post war period. This equilibrium price represents the effects of demographic shocks on the coalition of voters only on the housing market, and thus ignores other factors such as changes in the business cycle, technology and financial crises. Accordingly one should not expect it to closely track the actual house price that is affected by these non-demographic factors. The post war spike in the birth rate leads to a significant fall in the equilibrium house price starting in the 1970s as these households, who are mostly young renters, enter the housing market. As these households age and become homeowners they shift towards opposing the construction of new housing to protect the value of their stock of housing.

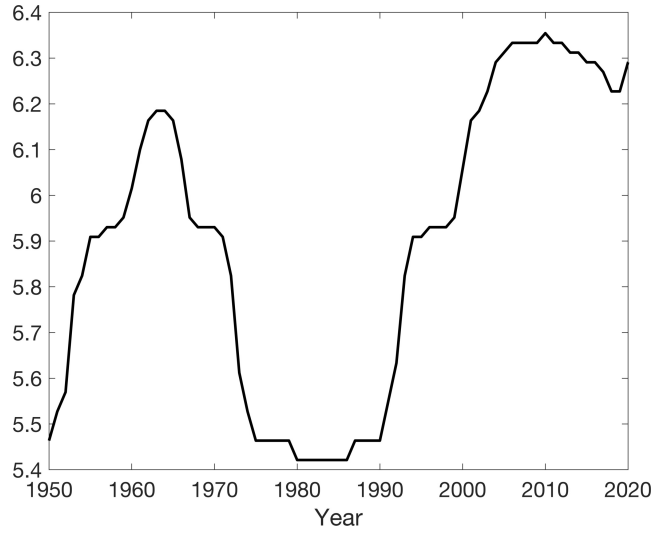


Figure 5: Steady State Equilibrium House Price

Using demographic projections out to 2100 we can estimate the impact the on-going aging of the US economy will have on the political economy of the US housing market as shown in figure 6. The IMF calculate demographic projections for the US population subject to three different assumptions about the level of immigration, of primarily young households, over the next 80 years. We find that for low to medium assumptions about immigration the aging of the US population will continue to increase the relative size of the electorate who support higher house prices - at least at the initial level of house prices. As a result, the equilibrium level of house prices increases over the next 80 years. For high level assumptions about immigration we find results similar to that of a spike in the birth rate, this is because immigrants tend to be younger and would change the demographic structure towards a lower median-age which would initially result in more political support for increase in the housing supply. The high immigration projection includes some long run cyclical dynamics as young immigrants eventually age and swing between the NIMBY and anti-NIMBY coalitions.

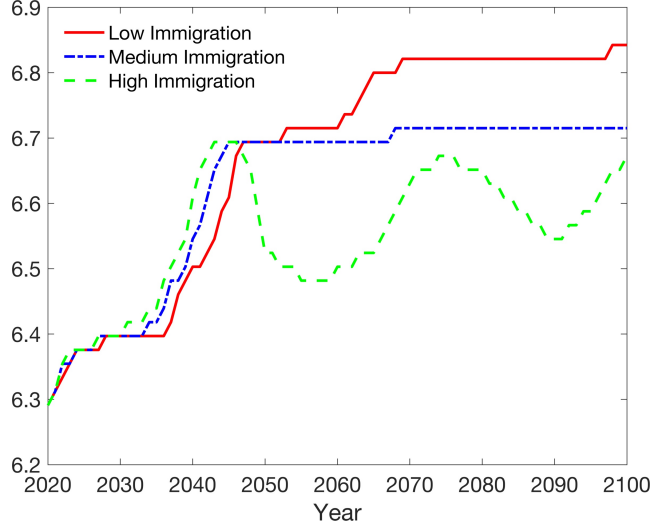


Figure 6: Steady State Equilibrium House Price Projections

5.2 Impulse Response Functions

In this next section we calculate the dynamics of a demographic shock to the housing market in the form of a temporary baby boom. We start the model in its equilibrium and then increase the birthrate of new households by 10 per cent for 10 years as shown in 7. After 25 years, when those who were born during the boom enter the labor and housing market, the effect of the spike in the birthrate starts to affect the demographic structure of households in the model. We then calculate the equilibrium response of the economy across the subsequent 80 years to see the effect of the baby boom through time.

In order to solve the model's dynamics we must take a stance on how households expectations of the future path of house prices are formed as expectations about the future capital gains or losses affect households demand for housing. To keep the model tractable we assume that households expect that house prices follow a random walk and thus in expectation will remain at their current level - at least with respect to the demographic changes in the model.

Initially, the generation that was born during the boom, decreases the median age of households within the economy as the number of young household increases. After 10 years the population boom ends and the birth rate returns to normal. After the birthrate returns to its steady state level, the generation born during the boom continues to age, pushing the average age above it's steady state level until they all die out 65 years later. This generation, being larger than the generations that come before or after them, have disproportionately more voting power. The impact of this electorally powerful cohort will vary depending on how the preferences of those born during the boom differ from the rest of the population.

When young, those born during the boom are disproportionately renters, with lower incomes and lower net wealth. As a result they use their outsized voting power to increase the supply

of new houses and push down the price of housing. The lower price of housing increases the homeownership rate across all ages, although the impact on the overall home ownership rate is small. In general equilibrium this downward pressure on the price of housing is partially offset by the creation of new home owners who on average prefer higher prices once they purchase a home.

The impact of this initial change in the housing market is to decrease both the level of across-generation inequality (as older and wealthier households relative net worth falls with house prices) and within generational inequality (as renters of all ages are able to either gain access to the housing market for the first time or alternatively pay a lower rate of rent).

However this impact is short lived. Greater political support for lower house prices ends after 20 years and prices then rise to above their steady state level. This reversal is driven by the increase in homeownership as the boom generation ages. The higher rate of homeownership and increases the number of households who expect to be net-sellers of housing and thus would prefer a higher price. This rise in home ownership is driven by two effects. The first is purely mechanical: homeownership rates increase with age as households build up their wealth and the generation born during the boom is comparatively large. Hence, we would expect an increase in the aggregate homeownership rate of the entire population as this outsized generation ages. The second effect is the endogenous impact of the vote to lower house prices in the first 20 years. This increased support for higher housing supply in the first two decades makes it easier for households to buy housing across *all* generations. Not only do those born during the boom have disproportionate voting power, but by initially supporting lower house prices they (unintentionally) increased the size of their voting coalition across generations by creating more homeowners among the households who are older.

For example, consider an older, low income household that was born 25 years before the baby boom. As a low income household they might optimally choose to rent for the entirety of their life. However, when those born during the boom starts to vote for lower house prices, their optimal decision may change allowing them to purchase housing for the first time. This change in their status from being a renter to a homeowner (who is unlikely to make additional purchases of housing) will lead them to switch their political support from lower to higher prices for the rest of their life.

This rise in house prices, however, leads to falls in the homeownership rates across all age levels. It also more then unwinds the decrease in generational inequality as the wealth ratio of 55-65 year olds to 25-35 year olds increases. The generation that is born after the baby boom face a housing market in which house prices are significantly higher. This leads them to have a lower rate of home ownership compared with the model's steady state. These high house prices persist for several decades those born during the boom start to die.

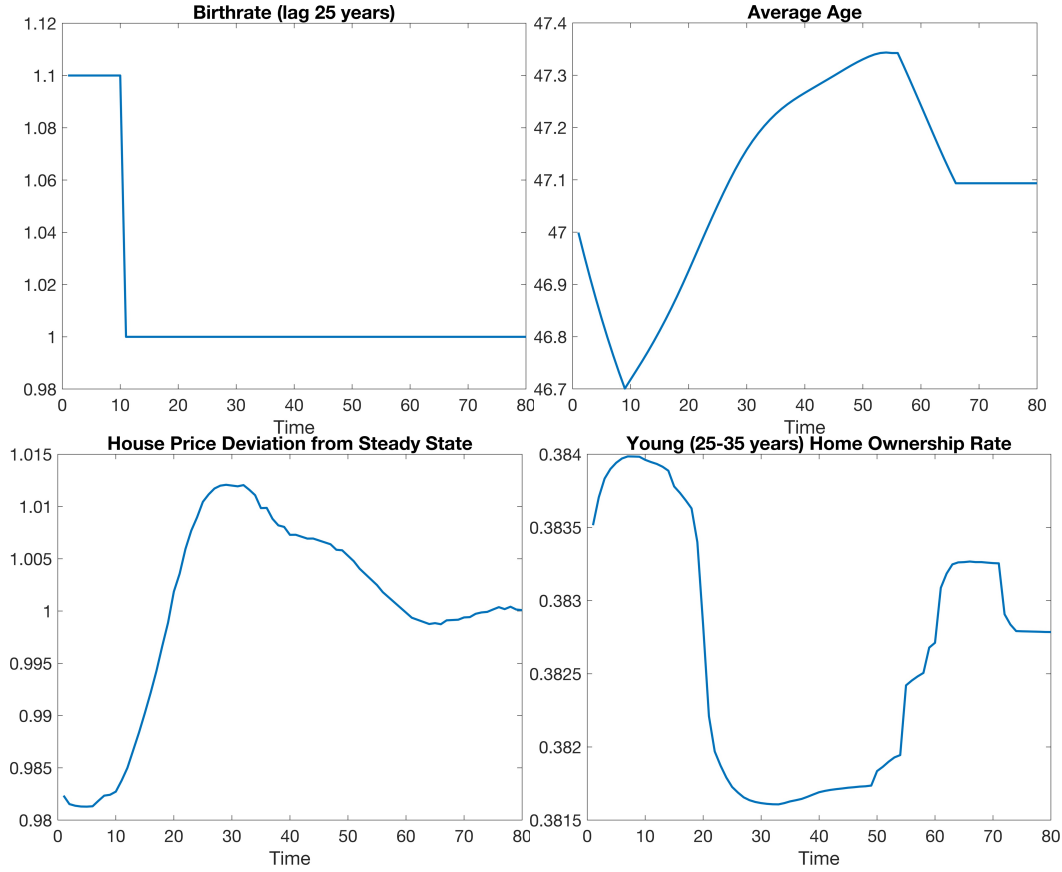


Figure 7: Impulse Response Functions for a Temporary Birthrate Increase

The impacts of the generation born during the baby boom can also be broken down by their effect on different cohorts as shown in figure 8. We analyze the effects of those born during the boom on their own 10 year long generation, their parents generation (who are 25 years older) and their children's generation (who are 25 years younger). Those born during the boom are generally able to use their outsized political power to increase their relative net-worth across the entire life cycle.

This demographic shock also affects the generations that come before and after the boom. The generation of parents that gave birth to those during the boom are initially worse off when house prices fall. However, they benefit towards the end of their life when their off-spring transition into homeownership and lift the equilibrium house price. The generation of children that came after are rendered permanently poorer due to the higher cost of housing, this leads them to have a permanently lower level of net worth across their entire life.

This mechanism outlines how the political economy mechanism of housing supply means that changes in the size of demographic structure can have significant effects on the housing market and economic outcomes for different generations.

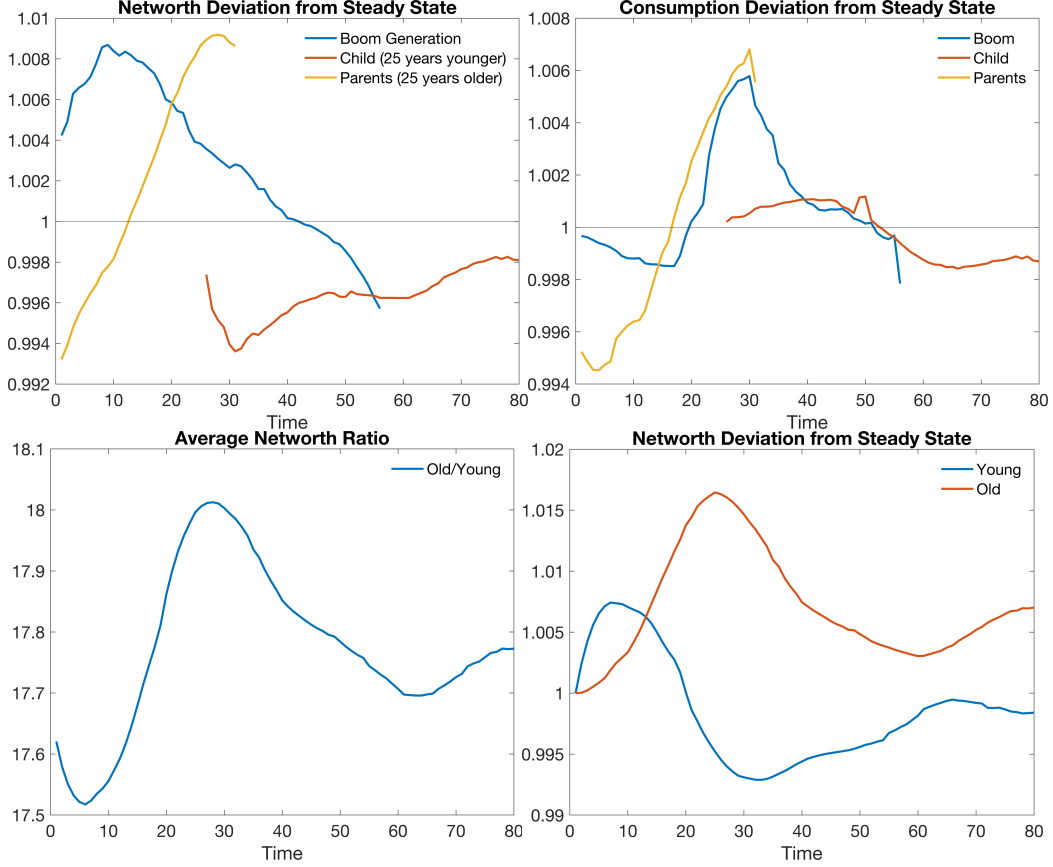


Figure 8: Impulse Response Functions for a Temporary Birthrate Increase

5.3 Permanent Demographic Shock

We next consider the impact of a permanent demographic shock as an alternative to a transitory shock. In this case the formation rate of new households falls 10 per cent forever. While the fall in the birth rate is permanent the effect on the model is temporary as the once new formation rate has time to filter through the demographic structure of the economy, the relative sizes of the different generations will return to the original steady state of the model. We find that this permanent reduction in the formation of new households thus causes a transitory increase in the average age that lasts 56 years. This increase in the average age increases the relative size of the older generations. Since these older generations are predominantly home owners and more likely to prefer a lower level of housing supply. Thus we see an increase in the equilibrium house price and a corresponding reduction in the homeownership rate and the consumption of housing.

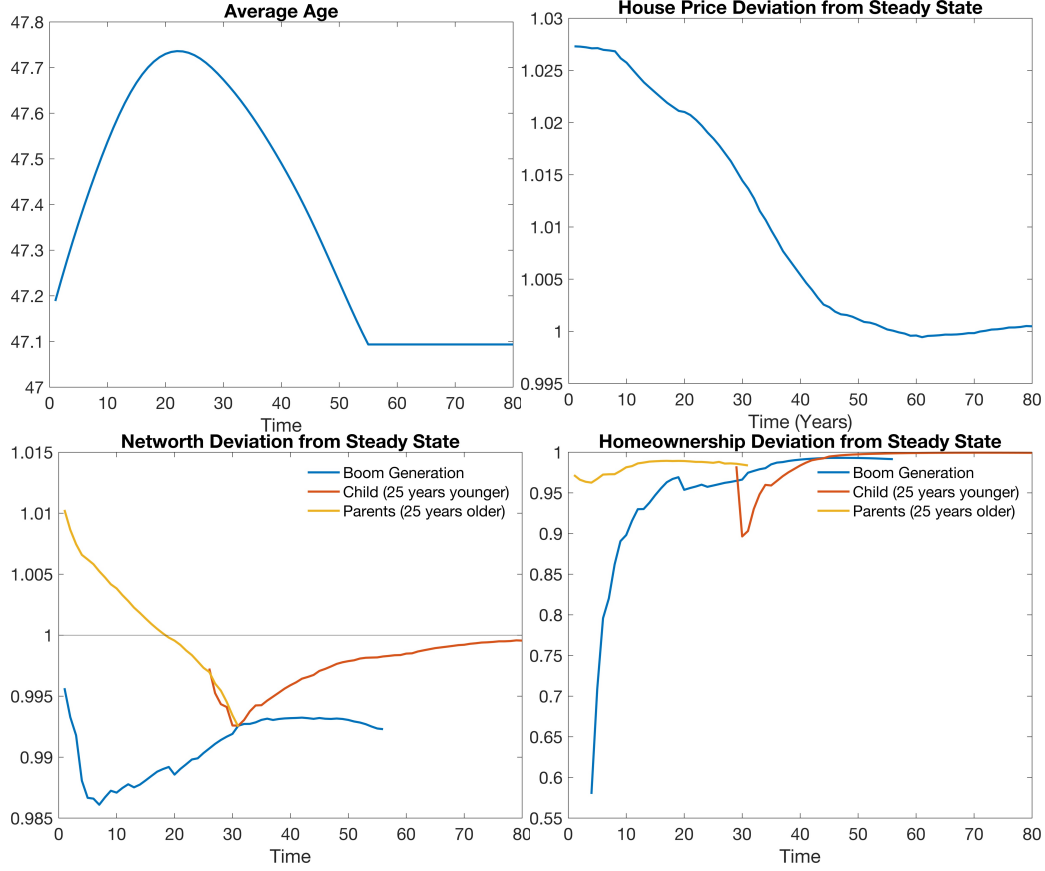


Figure 9: Impulse Response Function for Permanent Birthrate Fall

6 Robustness

We first consider how these results are affected by choice of calibration of the model. Varying the size of non-financial preferences over housing supply largely does not affect the dynamics of the model. While a stronger (weaker) non-financial preference for housing supply does lower (increase) the steady state house price to income ratio the impacts on the dynamics of the model are small. Regardless of whether the non-financial preferences are increased or decreased the impact of a temporary baby boom on house prices is broadly unchanged. This is because while the mean level of support changes under the alternative calibration differs the marginal impact of a boom generation is mostly unchanged.

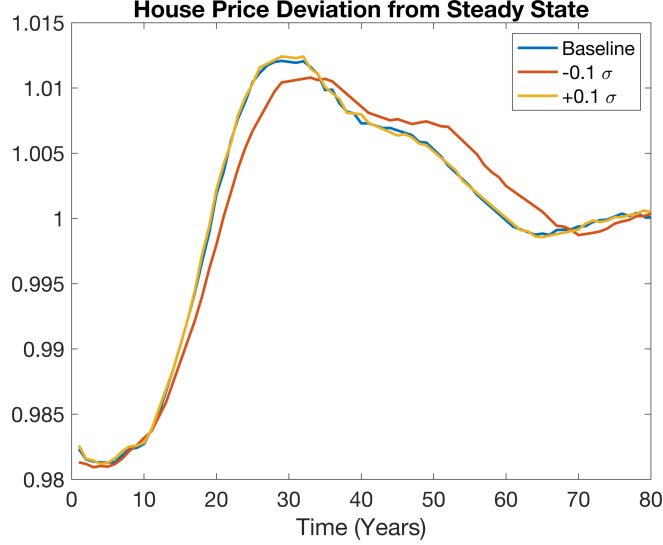


Figure 10: Steady State Equilibrium House Price

6.1 Alternative Voting Mechanisms

Finally we consider how variations in the voting mechanism may affect the results for the equilibrium house prices. We consider two alternatives: weighting household voting power by turnout rates and weighting households political power by the log of their net worth.

Weighting households vote by their turnout rate (in US elections) by age does not materially change the results. Younger households tend to have a lower voting rates, a result which has been constant across time. However, the overall estimated path of house prices does not change.

Alternatively, weighting by household's political power by using the log of their net worth, does materially change the path of projected house prices. The fall in house prices that is otherwise estimated to have occurred in the 1970s, when the baby boom generation are mostly renters, does not occur. Although they are more numerous than previous generations, their relatively low level of net worth means they a smaller degree of political leverage compared with the base case. As the economy ages, the increase in the price to income ratio is even larger as the aging population transition towards both home ownership and greater political power as their net worth increases. This alternative voting mechanism does not qualitatively change the overall results, but it does make it quantitatively larger in the 21st century.

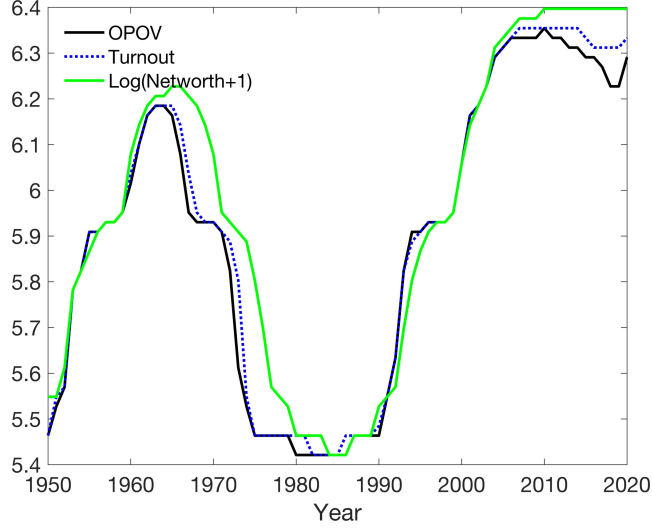


Figure 11: Steady State Equilibrium House Price

7 Conclusion

Local planning regulations and the political-economic constraints that bind them have long been defining features of housing markets across high-income countries. Given the importance of housing in homeowner's financial portfolios, it is natural that some households might seek to protect this value by voting in favor of regulations that restrict the supply of new housing. In this paper, we show how post-war demographic changes led to a shift in relative generation sizes and, consequently, changes in political support for additional housing supply. Our political-economic model of the housing market estimates that these demographic changes resulted in a peak-to-trough increase of house prices by 15 percent. Furthermore, future demographic projections indicate further increases in the house prices over the coming decades.

Unpacking the distributional impacts of this generational shift, we find that a one-off increase in the birth rate generally increases the relative consumption of the boom generation as they use their disproportionate voting power to improve their financial position throughout their life cycle. However, this comes at the cost of other generations: the consumption of the parent's generation initially falls during the boom, and the net worth of the baby boomers children is lower over their entire life cycle. This highlights how demographic shocks can have long-lasting effects, even after the baby boom generation has completely died off. Eventually, the housing market returns to equilibrium, but the effects of a baby boom can distort the housing market for both themselves and future generations.

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